

Grade 9 Mathematics

Factorization Lesson

1. Difference of Two Squares

The difference of two squares is an algebraic expression in the form $a^2 - b^2$. It can always be factorized into two brackets using the formula:

$$a^2 - b^2 = (a + b)(a - b)$$

Examples:

- $x^2 - 4 = (x + 2)(x - 2)$
- $x^2 - 16 = (x + 4)(x - 4)$
- $9x^2 - 1 = (3x + 1)(3x - 1)$
- $4x^2 - 25 = (2x + 5)(2x - 5)$
- $16a^2 - 49b^2 = (4a + 7b)(4a - 7b)$
- $x^2 - 9 = (x + 3)(x - 3)$
- $25m^2n^2 - 36 = (5mn + 6)(5mn - 6)$

2. Perfect Square Trinomials

A perfect square trinomial is an expression obtained by squaring a binomial. It follows one of the following patterns:

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

Examples:

- $x^2 + 6x + 9 = (x + 3)^2$
- $x^2 - 10x + 25 = (x - 5)^2$
- $4x^2 + 12x + 9 = (2x + 3)^2$
- $9x^2 - 30x + 25 = (3x - 5)^2$
- $16y^2 + 8y + 1 = (4y + 1)^2$
- $25a^2 - 40a + 16 = (5a - 4)^2$
- $36m^2 - 12m + 1 = (6m - 1)^2$

Key Reminder: Always check that the first and last terms are perfect squares and that the middle term matches twice the product of their square roots.